

# **Dysregulation of Glycemic Control**

Charles P. Scott, Ph.D.

October 6<sup>th</sup>, 2026

Modified from a presentation prepared by Dr. Peter Ronner on August 2<sup>nd</sup>, 2022

# Learning Objectives

1. Discuss the effect of type 1 diabetes on metabolism of glucose and fat, outline expected abnormalities in blood and urine laboratory values and suggest appropriate treatment.
2. Describe alterations to glucose counter-regulation in patients who have long-standing type 1 diabetes and explain why these alterations make patients more prone to hypoglycemia.
3. Summarize the abnormalities in endogenous glucose production, glycogen metabolism, and lipid metabolism that are often observed in patients who have type 2 diabetes.
4. Discuss the mechanisms and weaknesses of metformin, sulfonylurea/glinide drugs and GLP agonists as treatments for T2D.
5. Differentiate causes, diagnosis and treatment of post-prandial, fasting, ketotic and non-ketotic, hypoglycemia.

# Lesson Overview

## **Hyperglycemia**

Insulin deficiency (e.g., Type 1 diabetes)

Insulin resistance (e.g., Type 2 diabetes)

## **Hypoglycemia**

Ketotic

Non-ketotic

# Learning Objectives

1. Discuss the effect of type 1 diabetes on metabolism of glucose and fat, outline expected abnormalities in blood and urine laboratory values and suggest appropriate treatment.
2. Describe alterations to glucose counter-regulation in patients who have long-standing type 1 diabetes and explain why these alterations make patients more prone to hypoglycemia.
3. Summarize the abnormalities in endogenous glucose production, glycogen metabolism, and lipid metabolism that are often observed in patients who have type 2 diabetes.
4. Discuss the strengths and weaknesses of metformin, sulfonylurea/glinide drugs and GLP agonists as treatments for T2D.
5. Differentiate causes, diagnosis and treatment of post-prandial, fasting, ketotic and non-ketotic, hypoglycemia.

### **ScholarRx 3055.16**

A 12-year-old girl is brought to the emergency room because of vomiting. Her mother states that her daughter has had a severe case of the flu for the past week, has frequently been urinating, and, since this morning, has been unusually drowsy and fatigued. Laboratory studies show a blood glucose concentration of 480 mg/dL, and arterial pH of 7.0. Which of the following complications most likely explains this patient's condition?

- A. Increased glomerular capillary pressure
- B. Ketoacidosis
- C. Non-enzymatic glycation of proteins
- D. Osmotic diuresis
- E. Renal Hemorrhages
- F. Weight gain

# Type 1 Diabetes

Type 1 diabetes (T1D) results from autoimmune destruction of pancreatic  $\beta$ -cells

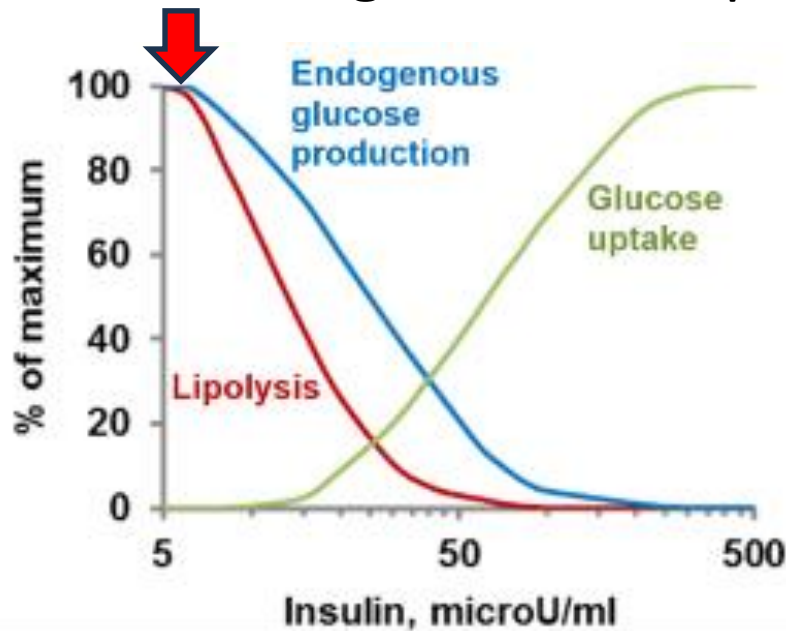
Patients with T1D produce little or no insulin

Insulin deficiency elevates risk for diabetic ketoacidosis (DKA)

# Diabetic Ketoacidosis (DKA)

Nearly pure and severe ketoacidosis develops only in the **absence or near-absence of insulin**

Insulin no longer inhibits lipolysis



High rate of fatty acid oxidation gives rise to excessive ketone body production

# Metabolism in absence of insulin

## Glucose production is maximal:

- No inhibition of glycogenolysis

- No inhibition of gluconeogenesis

## Lipolysis and fatty acid oxidation are near-maximal:

- No inhibition of lipolysis

- Ketone bodies are produced in excess of consumption

## Glucose use is minimal:

- No stimulation of GLUT-4 in muscle or adipose

- No stimulation of glycolysis

- No stimulation of glycogen synthesis

- Fatty acids are used in place of glucose

# Acid-base balance in DKA

Ketone bodies (acetoacetic acid &  $\beta$ -hydroxybutyrate) are released into blood

Acids dissociate,  $H^+$  lower pH

Bicarbonate +  $H^+$   $\rightarrow$   $CO_2$  (blown off)

$\rightarrow$  Increased respiratory rate

Kidneys cannot form as much bicarbonate as is used up

$\rightarrow$  Metabolic acidosis

(Low pH, low bicarb, low  $P_aCO_2$ )

# Dehydration in DKA

Glucose filtration compromised above  $\geq 180$  mg/dl

Ketone body filtration compromised above  $\geq 5$  mM

Water follows glucose and ketone bodies: polyuria



# Typical labs in DKA

|                               |            |
|-------------------------------|------------|
| Plasma glucose:               | >250 mg/dl |
| Bicarbonate:                  | < 15 mEq/l |
| pH of arterial blood:         | < 7.30     |
| Ketone bodies (serum, plasma) | ↑↑↑        |
| Anion gap: Elevated           | ↑          |

Treatment: INSULIN

### ScholarRx 3055.16

A 12-year-old girl is brought to the emergency room because of vomiting. Her mother states that her daughter has had a severe case of the flu for the past week, has frequently been urinating, and, since this morning, has been unusually drowsy and fatigued. Laboratory studies show a blood glucose concentration of 480 mg/dL, and arterial pH of 7.0. Which of the following complications most likely explains this patient's condition?

A. Increased glomerular capillary pressure

B. Ketoacidosis

C. Non-enzymatic glycation of proteins

D. Osmotic diuresis

E. Renal Hemorrhages

F. Weight gain

# Summary

In **diabetic ketoacidosis** glucose production is maximal, glucose use is minimal, lipolysis is near-maximal, and ketone body production far exceeds ketone body use, causing severe **acidosis** (happens in about 1 day). Treat with insulin.

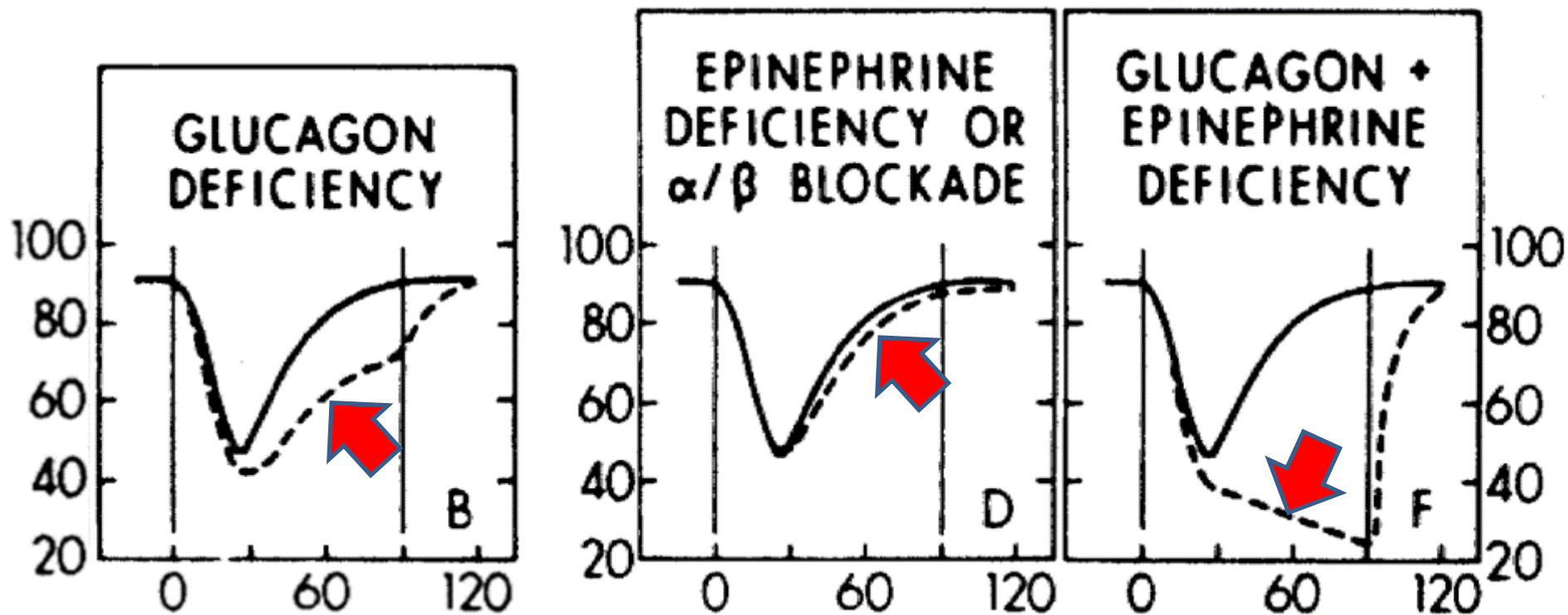
# Learning Objectives

1. Discuss the effect of type 1 diabetes on metabolism of glucose and fat, outline expected abnormalities in blood and urine laboratory values and suggest appropriate treatment.
2. Describe alterations to glucose counter-regulation in patients who have long-standing type 1 diabetes and explain why these alterations make patients more prone to hypoglycemia.
3. Summarize the abnormalities in endogenous glucose production, glycogen metabolism, and lipid metabolism that are often observed in patients who have type 2 diabetes.
4. Discuss the strengths and weaknesses of metformin, sulfonylurea/glinide drugs and GLP agonists as treatments for T2D.
5. Differentiate causes, diagnosis and treatment of post-prandial, fasting, ketotic and non-ketotic, hypoglycemia.

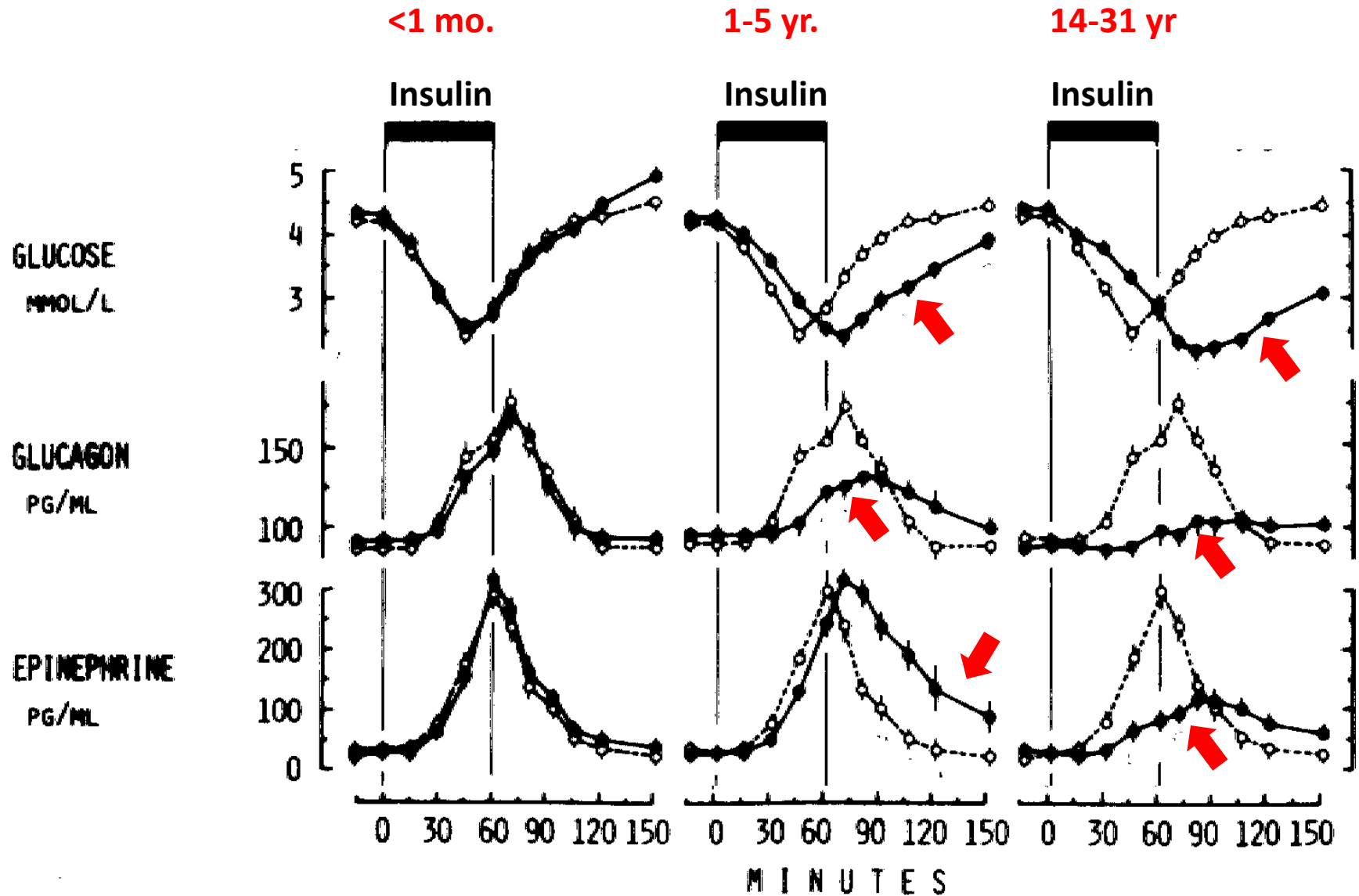
A 64-year-old patient is brought to the Emergency Department with a fracture following a fall in his kitchen. The patient doesn't remember falling. The patient was brought to the Emergency Department three months ago with trauma from an automobile accident after he blacked out while driving. The patient was diagnosed with type 1 diabetes at 13-years-old. His blood glucose at admission is 48 mg/dL. What is the most likely reason for this patient's hypoglycemia?

- A. Dementia
- B. Insulinoma
- C. Insulin resistance
- D. Loss of insulin counter-regulation
- E. Medication error

# Which counterregulatory hormones are essential?



# Lost counter-regulatory responses in T1D



A 64-year-old patient is brought to the Emergency Department with a fracture following a fall in his kitchen. The patient doesn't remember falling. The patient was brought to the Emergency Department three months prior with trauma from an automobile accident after he blacked out while driving. The patient was diagnosed with type 1 diabetes at 13-years-old. His blood glucose at admission is 48 mg/dL. What is the most likely reason for this patient's hypoglycemia?

- A. Dementia
- B. Insulinoma
- C. Insulin resistance
- D. Loss of insulin counter-regulation
- E. Medication error

# Conclusions

Longstanding diabetes is associated with a loss of the counter-regulatory response.

Hypoglycemia is a limiting factor when treating diabetes with insulin or drugs that increase insulin secretion.

Keep in mind that hypoglycemia is acutely dangerous.

# Learning Objectives

1. Discuss the effect of type 1 diabetes on metabolism of glucose and fat, outline expected abnormalities in blood and urine laboratory values and suggest appropriate treatment.
2. Describe alterations to glucose counter-regulation in patients who have long-standing type 1 diabetes and explain why these alterations make patients more prone to hypoglycemia.
3. **Summarize the abnormalities in endogenous glucose production, glycogen metabolism, and lipid metabolism that are often observed in patients who have type 2 diabetes.**
4. Discuss the strengths and weaknesses of metformin, sulfonylurea/glinide drugs and GLP agonists as treatments for T2D.
5. Differentiate causes, diagnosis and treatment of post-prandial, fasting, ketotic and non-ketotic, hypoglycemia.

### **ScholarRx 3845.20**

A 46-year-old man comes to his primary care physician because of increased thirst and a recent onset of frequent urination. He is 182 cm (6 ft) tall and weighs 107 kg (236 lb); body mass index is 32 kg/m<sup>2</sup>. Urinalysis shows 3+ glucose. Decrease in which of the following processes is most likely responsible for this patient's symptoms?

- A. Cellular glucose uptake
- B. Glucagon secretion
- C. Glucocorticoid secretion
- D. Glucose transporter 1 receptors
- E. Hypothalamic hormone synthesis

# Metabolic deficiencies in T2D

- Defective insulin production and/or insulin resistance
- Insufficient inhibition of endogenous glucose production (glycogenolysis, gluconeogenesis)
- Insufficient activation of glucose uptake and utilization
- Hyperglycemia
- **NOTE:** lipolysis is generally controlled sufficiently (i.e., DKA is not typically observed)

# Genetic risk of diabetes

Positive family history = excellent predictor

Type 2 diabetes susceptibility genes: not yet useful

But: there are a few monogenic forms of diabetes ..

# Monogenic forms of diabetes

**Persistent neonatal diabetes** (onset within 1<sup>st</sup> year of life)

**Transient neonatal diabetes** (onset within 1<sup>st</sup> year of life)

Both involve mutations to ATP-sensitive K<sup>+</sup> channel

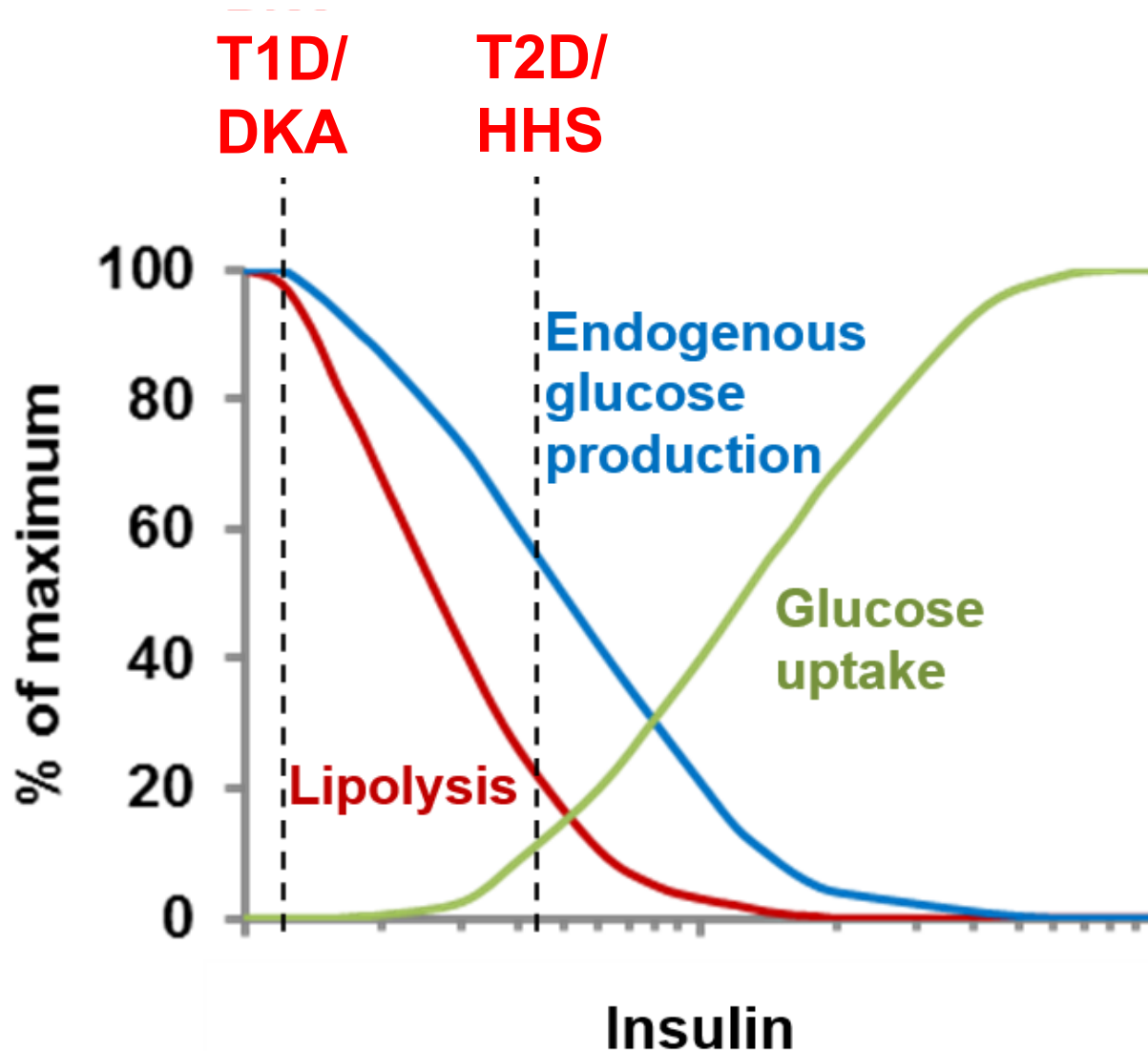
- Sulfonylureas & glinides often effective

**Maturity-onset diabetes of the young (MODY)**

- More genetically diverse (e.g., glucokinase mutation)

Genetic testing is indicated, influences Rx

# Insulin in HHS (vs. DKA)



# Hyperosmolar Hyperglycemic State

Requires increased counter-regulatory hormones & grossly impaired (not absent!) insulin secretion:

Not enough insulin to suppress endogenous glucose production

Enough insulin to suppress lipolysis and hence ketone body production

In stressed elderly pts. w/ long-standing type 2 diabetes

(Occasionally in new-onset, yet undiagnosed type 1 diabetes)

# HHS develops over several days

Excessive glucose production

Low glucose use

Glucose spills into urine and leads to **severe dehydration**

Dehydration ↓  $O_2$  perfusion, affecting mental status



# DKA ..... HHS

Classic:

Type 1 diabetes

Tell-tale differences:

pH <7.30

$\text{HCO}_3^-$  <15 mEq/l

Ketone bodies: ↑↑↑

Glucose >250 mg/dl

Osmolality <320  
mOsm/kg

Classic:

Type 2 diabetes

Tell-tale differences:

pH >7.30

$\text{HCO}_3^-$  >15 mEq/l

Ketone bodies: (↑)

Glucose >600 mg/dl

Osmolality >320  
mOsm/kg

# What brings patients to medical attention?

**DKA**

Acidosis



**HHS**

Dehydration

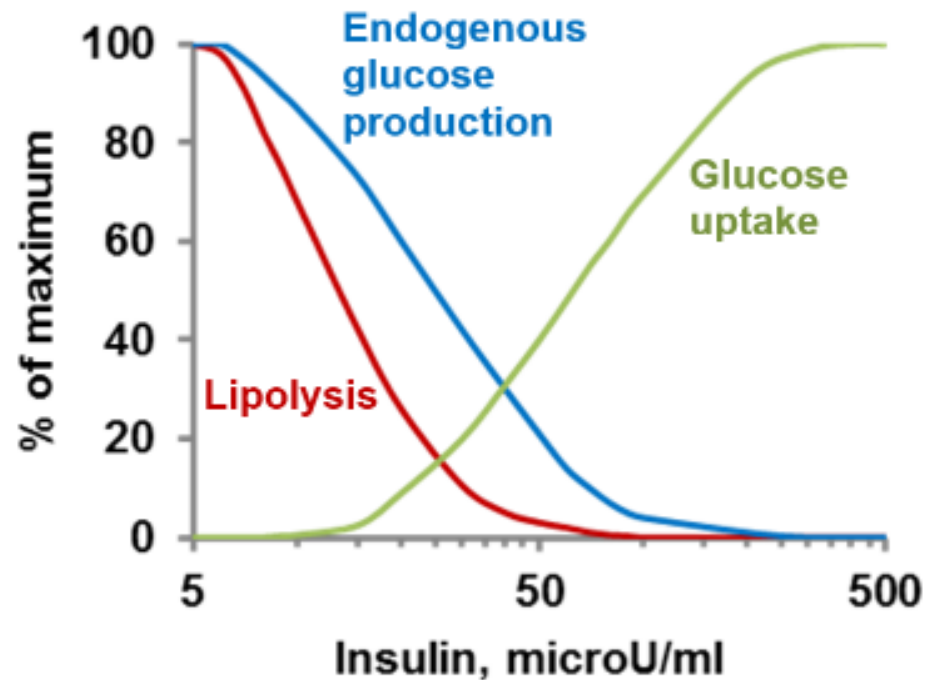


# DKA ..... HHS in the clinic

DKA



HHS



### **ScholarRx 3845.20**

A 46-year-old man comes to his primary care physician because of increased thirst and a recent onset of frequent urination. He is 182 cm (6 ft) tall and weighs 107 kg (236 lb); body mass index is 32 kg/m<sup>2</sup>. Urinalysis shows 3+ glucose. Decrease in which of the following processes is most likely responsible for this patient's symptoms?

- A. Cellular glucose uptake
- B. Glucagon secretion
- C. Glucocorticoid secretion
- D. Glucose transporter 1 receptors
- E. Hypothalamic hormone synthesis

# Summary of last topics continued

In **diabetic ketoacidosis** glucose production is maximal, glucose use is minimal, lipolysis is near-maximal, and ketone body production far exceeds ketone body use, causing severe **acidosis** (happens in about 1 day). Treat with insulin.

In **hyperosmolar hyperglycemic state**, glucose production is near maximal and glucose use is near minimal, but lipolysis and ketone body production are controlled. Glucosuria causes severe **dehydration** over a period of days. Treat for dehydration.

# Learning Objectives

1. Discuss the effect of type 1 diabetes on metabolism of glucose and fat, outline expected abnormalities in blood and urine laboratory values and suggest appropriate treatment.
2. Describe alterations to glucose counter-regulation in patients who have long-standing type 1 diabetes and explain why these alterations make patients more prone to hypoglycemia.
3. Summarize the abnormalities in endogenous glucose production, glycogen metabolism, and lipid metabolism that are often observed in patients who have type 2 diabetes.
- 4. Discuss the strengths and weaknesses of metformin, sulfonylurea/glinide drugs and GLP agonists as treatments for T2D.**
5. Differentiate causes, diagnosis and treatment of post-prandial, fasting, ketotic and non-ketotic, hypoglycemia.

## ScholarRx 1950.20

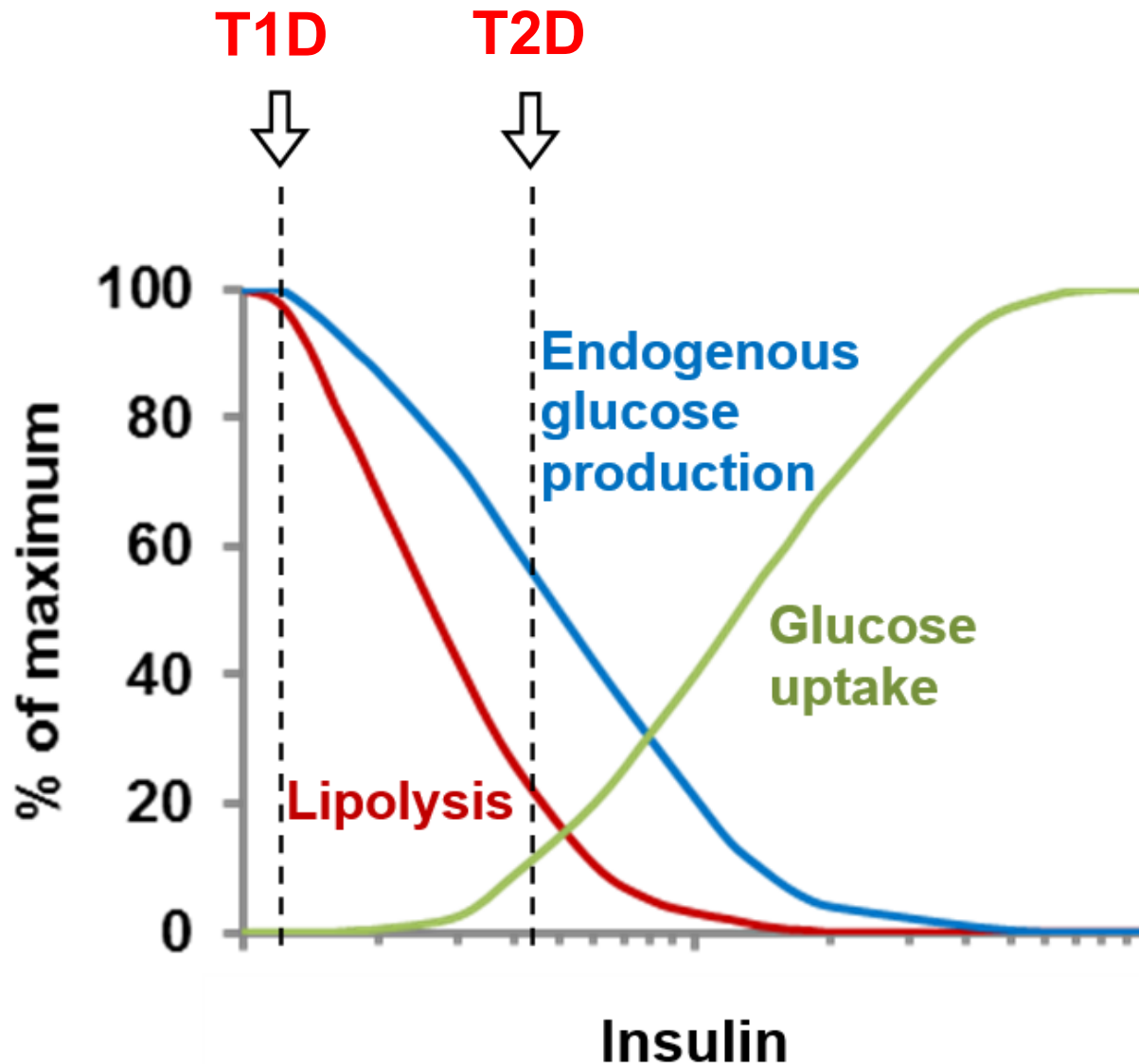
A 57-year-old man is brought to the emergency department after being found with an empty pill bottle at his bedside. He has diffuse abdominal pain. He has a history of depression and past suicide attempts. He also has a history of type 2 diabetes mellitus, currently managed with medication. His vital signs are stable. Laboratory studies show:

Na<sup>+</sup>: 140 mEq/L, K<sup>+</sup>: 4.8 mEq/L, Cl<sup>-</sup>: 100 mEq/L, HCO<sub>3</sub><sup>-</sup>: 18 mEq/L,  
Glucose: 110 mg/dL, BUN: 30 mg/dL, Serum creatinine: 3.5 mg/dL

Which of the following is the mechanism of the drug most likely causing this patient's findings?

- A. Decreases glucagon release
- B. Directly binds insulin receptors
- C. Inhibits gluconeogenesis
- D. Inhibits intestinal  $\alpha$ -glucosidases
- E. Triggers insulin release

# Insulin in T2D



# How is GNG regulated?

Long-term at 1<sup>st</sup> enzyme that catalyzes an “irreversible” reaction: PEPCK (OAA → PEP)

Stimulated by **epinephrine, glucagon**

Inhibited by insulin, AMP, AMPK

GNG is inhibited by the antidiabetic drug **metformin**

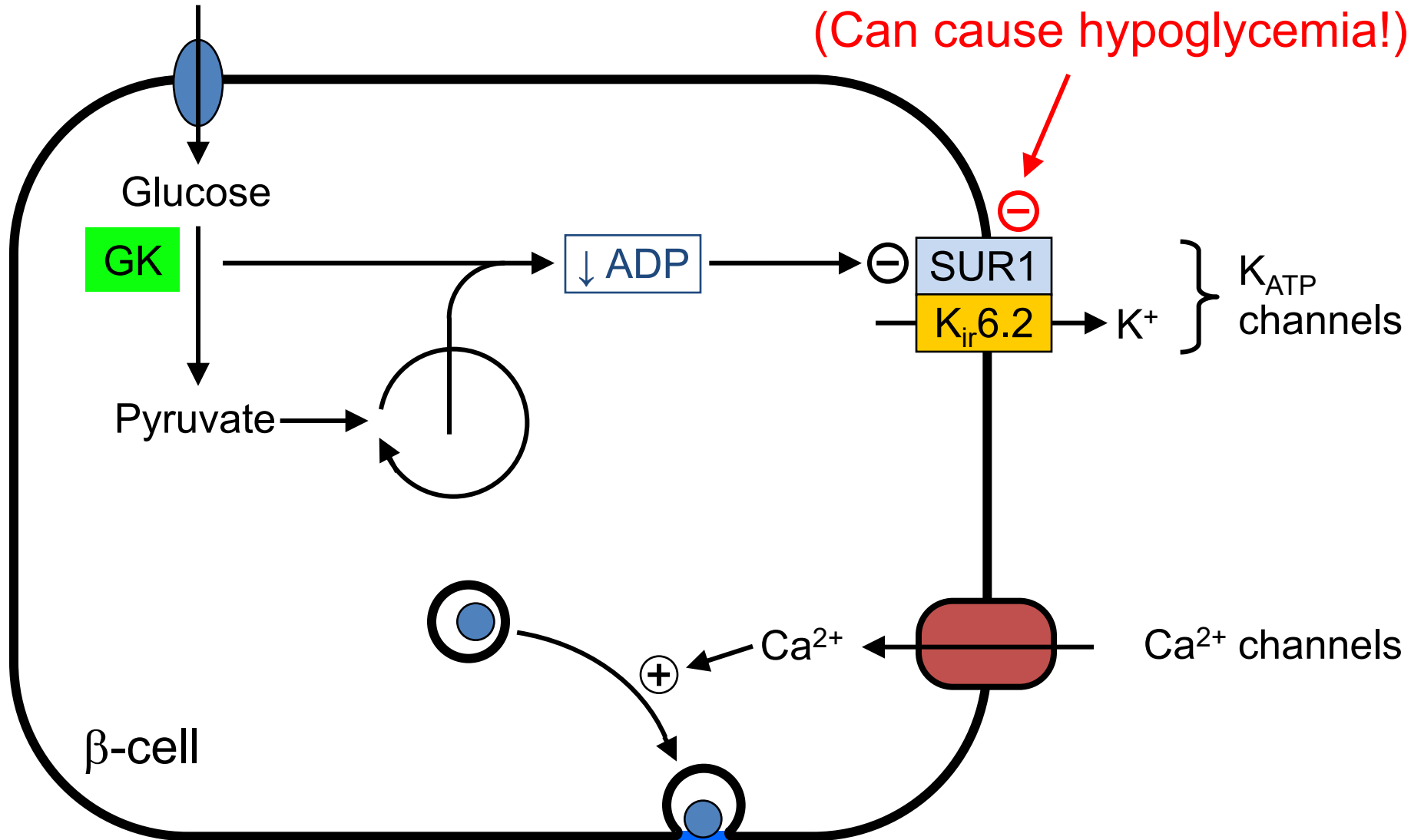
**Metformin** inhibits complex I in the mitochondria, which increases [AMP] and thereby stimulates AMPK

Metformin can cause lactic acidosis

Metformin is contraindicated in patients with poor renal function

# Sulfonylureas stimulate insulin secretion

Sulfonylureas, Glinides  
(Can cause hypoglycemia!)



# **GIP and GLPs enhance glucose-induced insulin secretion**

**GIP and GLPs stimulate insulin secretion only at [glucose]  
greater than about 110 mg/dl**

**Incretins delay gastric emptying and promote satiety**

**DPP-4 degrades GIP and GLPs**

**GIP receptor agonists, GLPs receptor agonists, and DPP-4  
inhibitors reduce [blood glucose] without causing  
hypoglycemia.**

**Incretins are contraindicated in patients with a history of  
pancreatitis. These drugs may also promote muscle  
wasting.**

## ScholarRx 1950.20

A 57-year-old man is brought to the emergency department after being found with an empty pill bottle at his bedside. He has diffuse abdominal pain. He has a history of depression and past suicide attempts. He also has a history of type 2 diabetes mellitus, currently managed with medication. His vital signs are stable. Laboratory studies show:

Na<sup>+</sup>: 140 mEq/L, K<sup>+</sup>: 4.8 mEq/L, Cl<sup>-</sup>: 100 mEq/L, HCO<sub>3</sub><sup>-</sup>: 18 mEq/L,  
Glucose: 110 mg/dL, BUN: 30 mg/dL, Serum creatinine: 3.5 mg/dL

Which of the following is the mechanism of the drug most likely causing this patient's findings?

- A. Decreases glucagon release
- B. Directly binds insulin receptors
- C. Inhibits gluconeogenesis
- D. Inhibits intestinal  $\alpha$ -glucosidases
- E. Triggers insulin release

# Learning Objectives

1. Discuss the effect of type 1 diabetes on metabolism of glucose and fat, outline expected abnormalities in blood and urine laboratory values and suggest appropriate treatment.
2. Describe alterations to glucose counter-regulation in patients who have long-standing type 1 diabetes and explain why these alterations make patients more prone to hypoglycemia.
3. Summarize the abnormalities in endogenous glucose production, glycogen metabolism, and lipid metabolism that are often observed in patients who have type 2 diabetes.
4. Discuss the strengths and weaknesses of metformin, sulfonylurea/glinide drugs and GLP agonists as treatments for T2D.
5. Differentiate causes, diagnosis and treatment of post-prandial, fasting, ketotic and non-ketotic, hypoglycemia.

### ScholarRx 3995.17

A 6-month-old breastfed girl is brought to the emergency department with seizures. Her mother had an uncomplicated term pregnancy. The patient is tachycardic and tachypneic, and heelstick reveals a glucose level of 55 mg/dL. Physical examination reveals hepatomegaly. Venous lactic acid level is 6 mmol/L (normal: <1 mmol/L). A deficiency in which of the following is most likely responsible for this patient's presentation?

- A.  $\alpha$ -1,4-glucosidase
- B.  $\alpha$ -1,6-glucosidase
- C. Fructokinase
- D. Galactokinase
- E. Galactose-1-phosphate uridylyltransferase
- F. Glucose-6-phosphatase

# Signs and Symptoms of Hypoglycemia

Nervousness, anxiety, irritability, lightheadedness,  
dizziness, headache

Sweating

Impaired vision

Confusion

Seizures

Loss of consciousness

Diabetes is a strong predictor of hypoglycemia

# Confirming Hypoglycemia Diagnosis



Hypoglycemia = blood glucose  $\leq$  70 mg/dl

**NOTE:** point of care glucose monitors are relatively insensitive to low [glucose]

# Causes of Hypoglycemia

Hypoglycemia is most often due to excess insulin

- injected (exogenous) insulin
- endogenous insulin (e.g., insulinoma)

Hypoglycemia can also be caused by defective gluconeogenesis

- congenital (e.g., von Gierke disease)
- defective energy metabolism (e.g., MCADD)
- Galactosemia, HFI

# What can be done about hypoglycemia?



Glucose (oral or I.V.)



Glucagon (stimulates glycogenolysis in liver but also causes nausea)

# Diagnosing Hypoglycemia

Postprandial hypoglycemia: mixed meal test



Fasting hypoglycemia: fast until hypoglycemic,  
stop after 72 h



# When does hypoglycemia occur?

**Postprandially**, due to

Hereditary fructose intolerance

Galactosemia

Gastric bypass surgery

In **fasting** state, due to

Insulin excess (e.g., insulinoma)

Deficient gluconeogenesis

Glycogen storage disease (alleviated by gluconeogenesis)

# Hypoglycemia w/ or w/o ketosis

Measure ketone bodies

Ketotic hypoglycemia

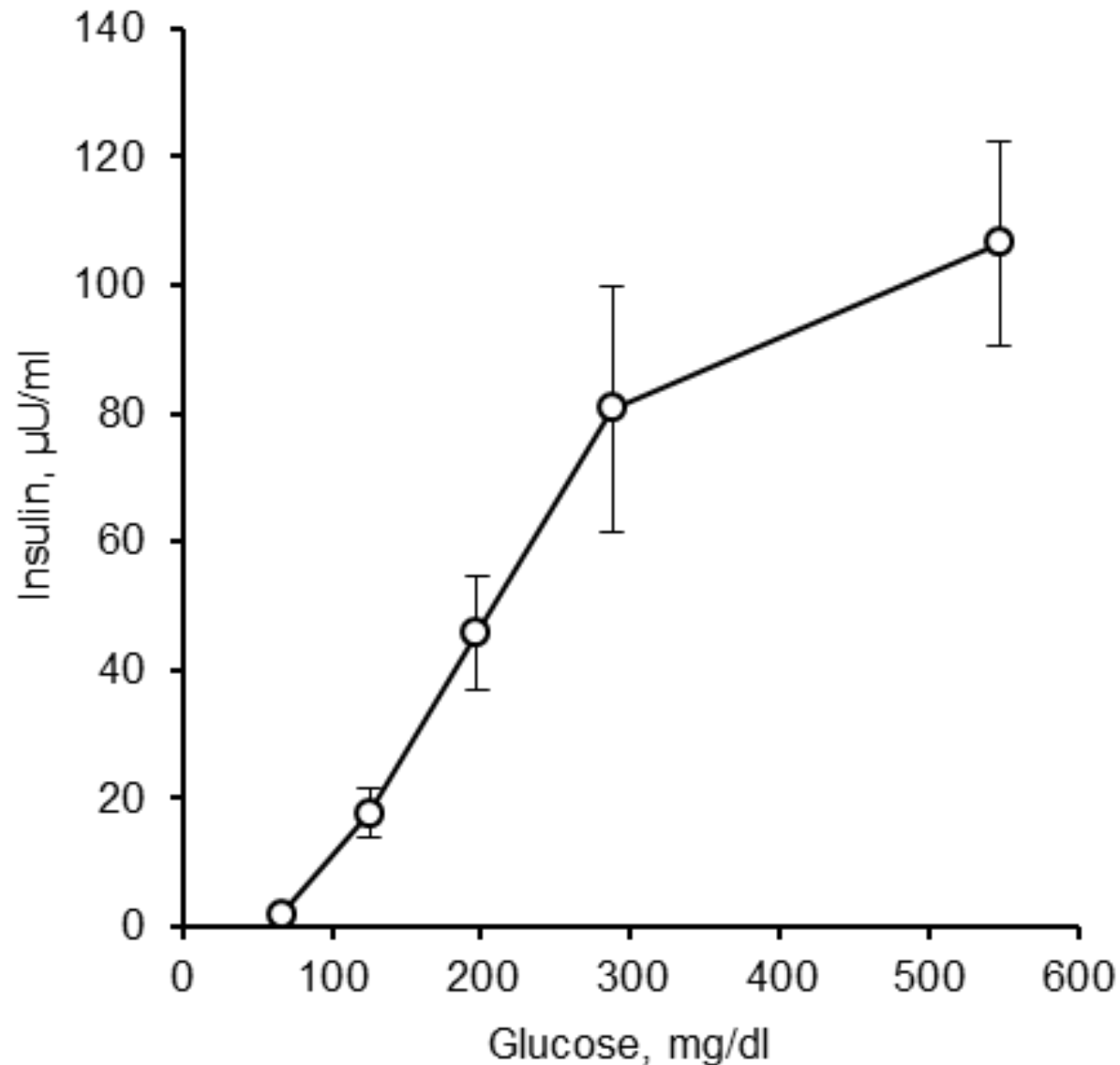
Non-ketotic hypoglycemia

# Ketotic Hypoglycemia

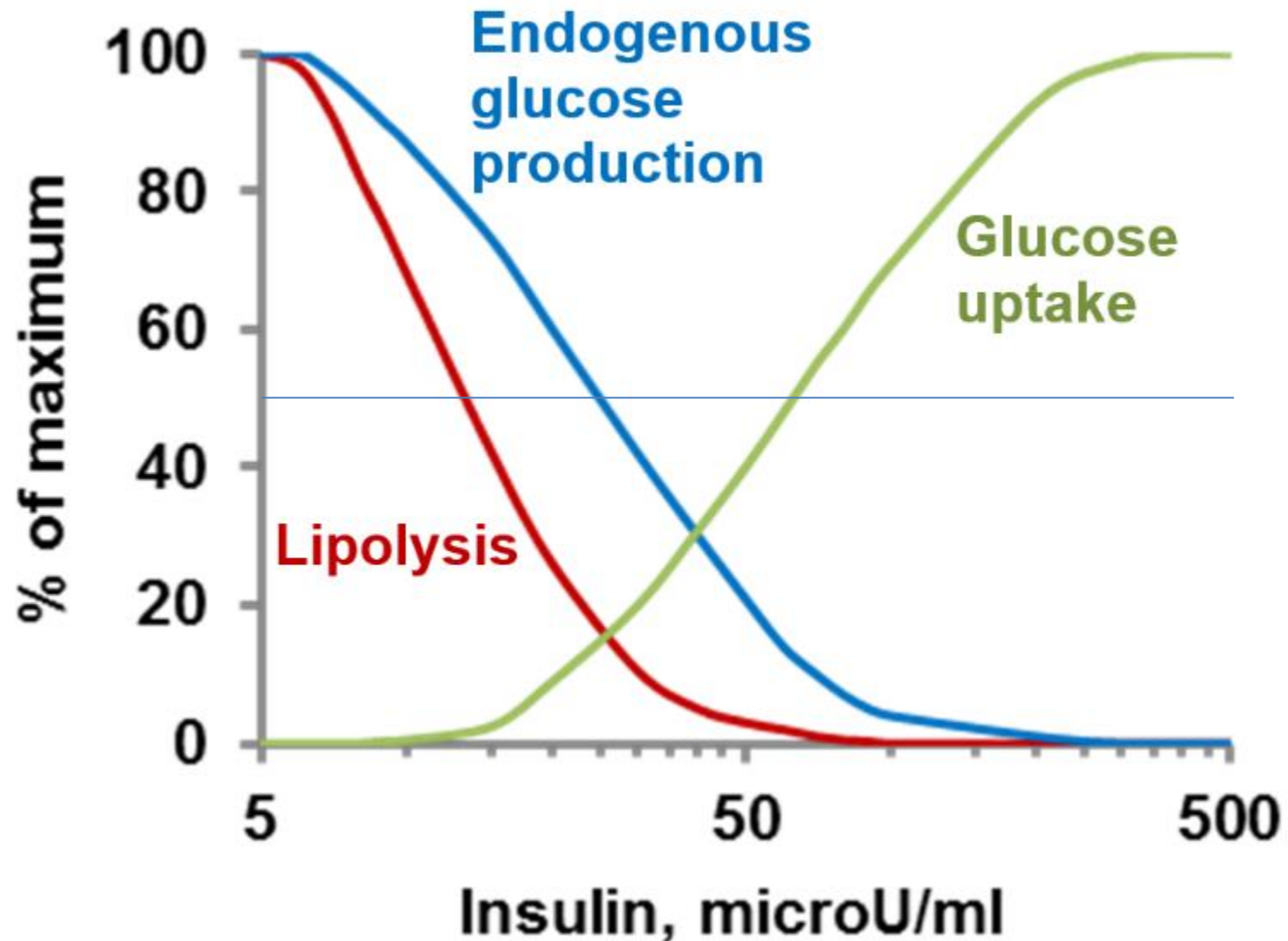
A. Deficient gluconeogenesis

**Note:** Glycogen storage disease usually causes only mild hypoglycemia, as long as gluconeogenesis can take over

# Glucose stimulates insulin secretion



# Insulin sensitivity of pathways of metabolism



# Deficiency in gluconeogenesis

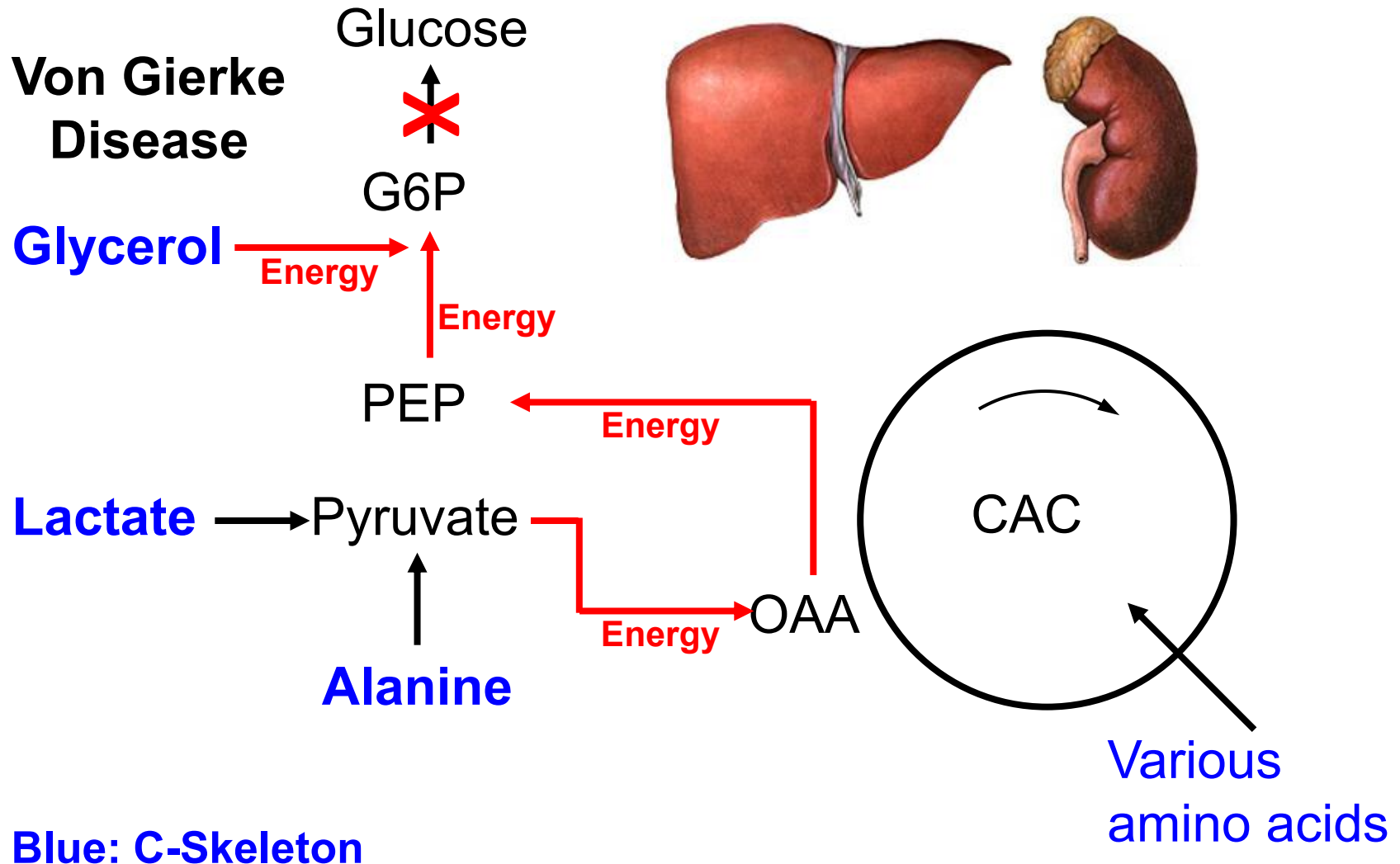
Hypoglycemia during prolonged fasting

Lactic acidosis during fasting

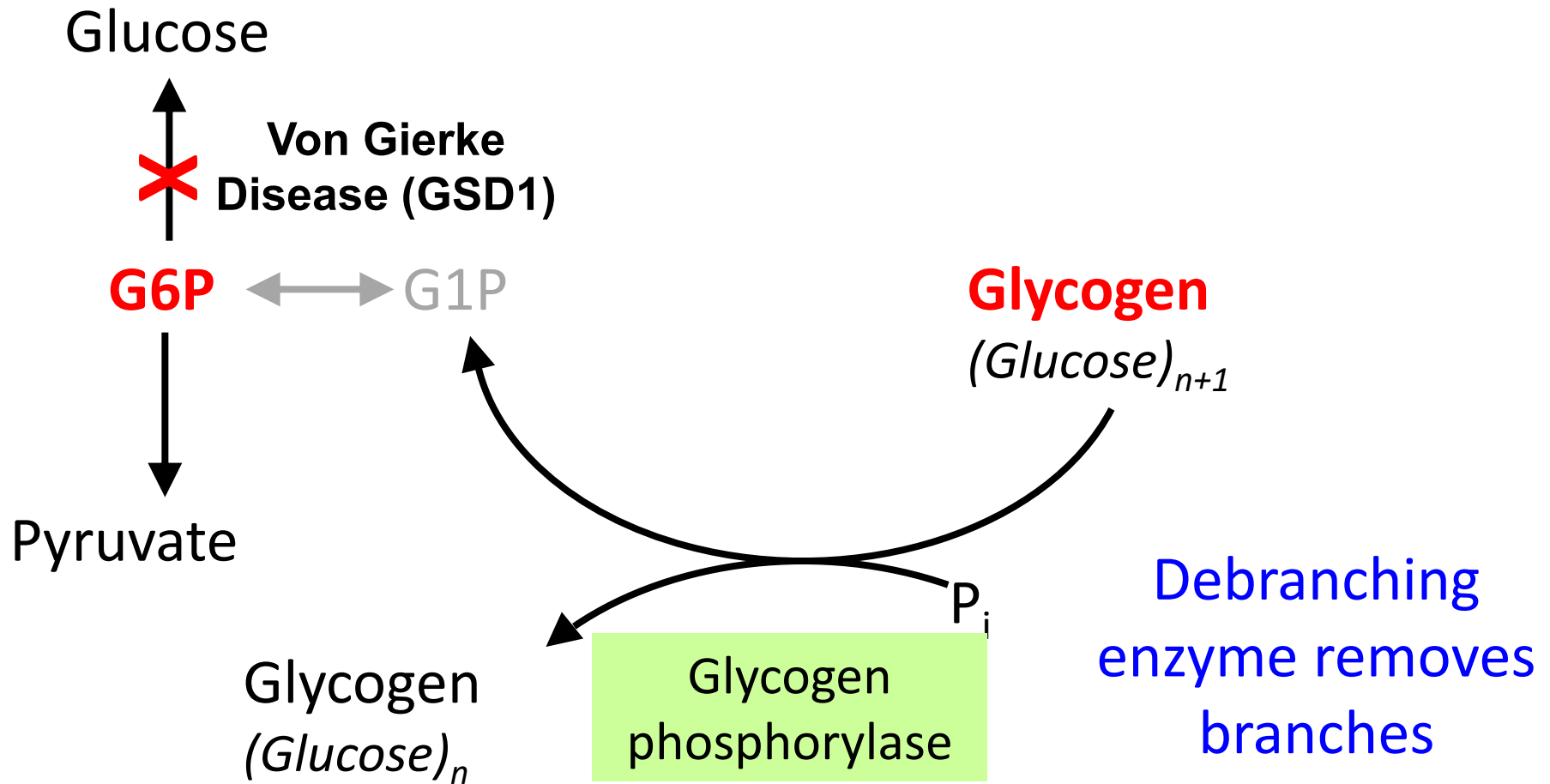
Example:

von Gierke disease (G6Pase deficiency)

# von Gierke Disease Inhibits Release of Glucose from Gluconeogenesis



# von Gierke Disease Inhibits Release of Glucose from Glycogen Degradation



# Deficiency in gluconeogenesis

Hypoglycemia during prolonged fasting

Lactic acidosis during fasting

Example:

von Gierke disease (G6Pase deficiency)

Avoid fasting

Regimen of uncooked corn starch

Low-carb diet & carb counting, no fruit, no milk

# Non-Ketotic Hypoglycemia

## A. Excess insulin

Excess injected insulin

Excess sulfonylurea or glinide

Insulinoma

Congenital hypersecretion of insulin

} Distinguish  
with C-peptide

## B. Deficient fatty acid oxidation (also impairs gluconeogenesis)

### ScholarRx 3995.17

A 6-month-old breastfed girl is brought to the emergency department with seizures. Her mother had an uncomplicated term pregnancy. The patient is tachycardic and tachypneic, and heelstick reveals a glucose level of 55 mg/dL. Physical examination reveals hepatomegaly. Venous lactic acid level is 6 mmol/L (normal: <1 mmol/L). A deficiency in which of the following is most likely responsible for this patient's presentation?

- A.  $\alpha$ -1,4-glucosidase
- B.  $\alpha$ -1,6-glucosidase
- C. Fructokinase
- D. Galactokinase
- E. Galactose-1-phosphate uridylyltransferase
- F. Glucose-6-phosphatase

# Summary

- T1D results from autoimmune destruction of pancreatic  $\beta$ -cells, leading to loss of insulin production
- Untreated T1D causes DKA through uncontrolled lipolysis
- Counterregulatory hormone control is lost in long-standing T1D, which increases risk for hypoglycemia
- T2D results from reduced insulin production and/or insulin resistance, leading to loss of inhibition of endogenous glucose production
- T2D is treated with inhibitors of gluconeogenesis (e.g., metformin) and/or drugs that stimulate insulin release (sulfonylureas, glinides, incretins)
- Drugs that act on the ATP-sensitive potassium channel (e.g., sulfonylureas & glinides) increase risk for hypoglycemia
- Hypoglycemia is diagnosed based on effects of feeding, fasting, and by the degree of ketosis